Bounded Relative Boundary Implies Narrow DNF Approximation

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Abstract

We prove Friedgut's conjecture (*J. Amer. Math. Soc.* 12 (1999)) that every increasing family in the p-biased discrete cube with bounded relative boundary can be approximated arbitrarily well by one whose minimal elements have size bounded independently of the dimension and the bias. For $0 < p \leq 1/2$, every increasing Boolean function with total resampling influence at most K is ε-close under μ_p^n to a monotone DNF of width $\exp(O((K+1)^2/\varepsilon^2))$, from which the full conjecture follows for all $p \in (0,1)$. We build on Hatami's pseudo-junta theorem (*Ann. of Math.* 176 (2012)), whose approximator need not be increasing. Our randomized shifts produce an increasing approximator that is measurable with respect to a controlled refinement of Hatami's adaptive partition. Retaining only short positive certificates yields the desired narrow DNF.

1 Introduction

The analysis of Boolean functions studies how local behavior on the discrete cube constrains global structure. Foundational results connect local sensitivity to sharp thresholds and low-complexity approximation [KKL88, FK96, Fri98]. A central theme is that a function that rarely changes under local perturbations should admit a simple structural description.

Let $\{0,1\}^n$ denote the discrete cube and write $[n] = \{1, \dots, n\}$. For $p \in (0,1)$, let μ_p be the Bernoulli measure on $\{0,1\}$ with $\mu_p(1) = p$, and let μ_p^n be its product measure on $\{0,1\}^n$. We write $\mathbb{P}_p$, $\mathbb{E}_p$, and Var_p for probability, expectation, and variance under μ_p^n, using the same notation for its coordinate marginals. The Hamming weight of $x \in \{0,1\}^n$ is $|x| = \sum_{i=1}^n x_i$. A Boolean function $f : \{0,1\}^n \to \{0,1\}$ is *increasing* if $x \le y$ coordinatewise implies $f(x) \le f(y)$. If $x^{\oplus i}$ is obtained by flipping coordinate i, define

$$\mathrm{sens}_f(x) = \big|\{i \in [n] : f(x) \neq f(x^{\oplus i})\}\big|.$$

For $X \sim \mu_p^n$, let $X^{(i)}$ be obtained by independently resampling coordinate i. We use the total resampling influence

$$I_p(f) = \sum_{i=1}^n \mathbb{P}_p[f(X) \neq f(X^{(i)})].$$

This is Hatami's resampling convention [Hat12, eq. (9)]. In the p-biased orthonormal Fourier basis, $I_p(f) = 2\sum_{S \subseteq [n]} |S|\, \widehat{f}(S)^2$.

A *k-junta* is a Boolean function that depends on at most k coordinates. When p is bounded away from zero and one, Friedgut's junta theorem states that bounded total influence implies approximation by a k-junta for some k independent of n [Fri98]. Such a bound on k cannot hold uniformly as p tends to zero. At $p = 1/n$, for example, the function $\mathrm{OR}_n(x) = x_1 \vee \cdots \vee x_n$ satisfies $I_{1/n}(\mathrm{OR}_n) = 2(1-1/n)^n = O(1)$ but remains a constant distance under $\mu_{1/n}^n$ from every k-junta with fixed k. Yet one can certify that $\mathrm{OR}_n(x) = 1$ by revealing a single coordinate equal to one. Sparse random graphs exhibit the same distinction. A triangle may occur anywhere in the graph, but its three edges certify its presence. Such local witnesses play a central role in threshold phenomena for random graphs and random constraint-satisfaction problems [FK96, Fri99].

For an increasing Boolean function f, the Margulis–Russo identity [Mar74, Rus81] gives

$$I_p(f) = 2p(1-p)\mathbb{E}_p[\mathrm{sens}_f(X)] = 2p(1-p)\frac{d}{dp}\mathbb{E}_p[f]. \tag{1}$$

Thus Friedgut's quantity

$$p\mathbb{E}_p[\mathrm{sens}_f(X)] = \frac{I_p(f)}{2(1-p)} = \frac{d}{d\log p}\mathbb{E}_p[f]$$

measures the derivative of the acceptance probability $\mathbb{E}_p[f] = \mathbb{P}_p[f = 1]$ on the multiplicative rather than additive scale of p. We therefore refer to $p\mathbb{E}_p[\mathrm{sens}_f(X)]$ as the relative boundary.

Friedgut's conjecture asks whether every increasing function with bounded relative boundary can be approximated by a function with short positive witnesses, as in the example of OR_n. For $T \subseteq [n]$, a *positive term* is the conjunction $\bigwedge_{i \in T} x_i$. A *monotone DNF* is a disjunction of positive terms, and its width is the largest $|T|$ among them. Each term is a local positive witness because setting its coordinates to one forces acceptance. We use the usual conventions that the empty conjunction is one, the empty disjunction is zero, and both constant functions have width zero. For a statement P, write $\mathbf{1}[P]$ for its indicator. For $\mathcal{B} \subseteq \{0,1\}^n$, write $\mathbf{1}_{\mathcal{B}}(x) = \mathbf{1}[x \in \mathcal{B}]$, and call $\mathcal{B}$ increasing when $\mathbf{1}_{\mathcal{B}}$

is increasing. An element $z \in \mathcal{B}$ is minimal if no distinct $y \in \mathcal{B}$ satisfies $y \le z$. If $\mathcal{B}$ is increasing, the terms $\bigwedge_{i:z_i=1} x_i$ indexed by its minimal elements form the canonical monotone DNF of $\mathbf{1}_{\mathcal{B}}$. Thus $\mathbf{1}_{\mathcal{B}}$ has a monotone DNF of width at most k if and only if every minimal element has Hamming weight at most k. For families $\mathcal{A}, \mathcal{B} \subseteq \{0,1\}^n$, write $\mathcal{A} \triangle \mathcal{B}$ for their symmetric difference.

Conjecture 1.1 (Friedgut's conjecture)**.** *For every $c > 0$ and $0 < \varepsilon < 1$, there is $k = k(c, \varepsilon) < \infty$ such that the following holds for every $p \in (0,1)$ and $n \ge 1$. If $\mathcal{A} \subseteq \{0,1\}^n$ is increasing and*

$$p\mathbb{E}_p[\mathrm{sens}_{\mathbf{1}_{\mathcal{A}}}(X)] \le c,$$

then there is an increasing $\mathcal{B} \subseteq \{0,1\}^n$ satisfying $\mu_p^n(\mathcal{A} \triangle \mathcal{B}) \le \varepsilon$ whose minimal elements all have Hamming weight at most k.

Our main theorem gives an explicit width bound in the low-bias regime.

Theorem 1.2 (Narrow-DNF approximation)**.** *There is an absolute constant $C > 0$ such that the following holds. Let $n \ge 1$, $K > 0$, $0 < p \le 1/2$, and $0 < \varepsilon < 1$. If an increasing function $f : \{0,1\}^n \to \{0,1\}$ satisfies $I_p(f) \le K$, then there is a monotone DNF g of width at most $\exp(C(K+1)^2/\varepsilon^2)$ such that*

$$\mathbb{P}_p[f(X) \ne g(X)] \le \varepsilon.$$

Together with the high-bias reduction, Theorem 1.2 proves Friedgut's conjecture [Fri99, Conjecture 1.5] for all $p \in (0,1)$. Our result shows that narrow monotone DNFs, rather than ordinary juntas, provide the appropriate dimension-free approximating class for sparse product spaces.

The gap after Hatami. Hatami's structure theorem replaces an ordinary junta by a *pseudo-junta* [Hat12]. For $S \subseteq [n]$, write x_S for the restriction of x to S. Hatami supplies a family $\mathcal{J} = (J_S)_{S \subseteq [n]}$ of local activation functions $J_S : \{0,1\}^S \to \{0,1\}$ and an approximator h of f. On input x, the activations determine the information

$$A_{\mathcal{J}}(x) = \bigcup_{S : J_S(x_S)=1} S, \qquad \Phi_{\mathcal{J}}(x) = (A_{\mathcal{J}}(x), x_{A_{\mathcal{J}}(x)}). \tag{2}$$

Thus $\Phi_{\mathcal{J}}(x)$ records the coordinates activated at x together with their values. We call two inputs $x, y \in \{0,1\}^n$ members of the same *adaptive atom* if $\Phi_{\mathcal{J}}(x) = \Phi_{\mathcal{J}}(y)$, and these atoms form the adaptive partition of $\{0,1\}^n$. We identify an atom with the corresponding value of $\Phi_{\mathcal{J}}$ when convenient. The value of h depends only on the information recorded by $\Phi_{\mathcal{J}}$. Equivalently, there is a readout map Γ such that $h(x) = \Gamma(\Phi_{\mathcal{J}}(x))$, so h is constant on every adaptive atom. We refer to this property as measurability with respect to the adaptive partition. For $X \sim \mu_p^n$, Hatami's representation satisfies $\mathbb{P}_p[f(X) \ne h(X)] = O(\varepsilon)$ and $\mathbb{E}_p|A_{\mathcal{J}}(X)| = O_{K,\varepsilon}(1)$. When $p \le 1/2$, the activations in Hatami's construction are increasing. The approximator h, however, need not be increasing, even when the target f is. For a general family of increasing activations, the best increasing approximator measurable on the original partition can have larger error than the best unrestricted approximator on that partition (Remark 3.3).

Proof architecture. Starting from f, we first obtain Hatami's approximator h, then construct an increasing adaptive approximator u, and finally extract a monotone DNF g. Our proof proceeds through the following sequence of transformations.

$$f \xrightarrow[\text{Hatami}]{\text{adaptive structure}} h \xrightarrow[\text{this paper}]{\text{controlled monotonicization}} u \xrightarrow[\text{this paper}]{\text{positive certificates}} g.$$

We extract and track two quantitative bounds from Hatami's activation family. Every nonzero activation examines at most $d = O((K+1)/\varepsilon)$ coordinates. We also introduce the *multiplicity-counted activation load*

$$\Lambda_p(\mathcal{J}) = \sum_{S \subseteq [n]} |S| \mathbb{P}_p[J_S(X_S) = 1] \tag{3}$$

and show directly from Hatami's construction that it is at most $\exp(O((K+1)^2/\varepsilon^2))$. The load bounds the expected size of the active union by counting overlapping sets with multiplicity. Each summand depends on at most d coordinates, so we can bound the effect of random forcing on each summand separately.

For the second arrow, we process the coordinates one at a time. At each coordinate, after fixing all other coordinates, we inspect the pair of values of the current approximator at its zero and one endpoints. The only pair that violates monotonicity is 10. An upper shift changes this pair to 11, while a lower shift changes it to 00. We choose the upper shift with probability $1-p$ and the lower shift with probability p. The randomness is only in the choice between these two deterministic transformations; the input itself is not resampled. Since the zero and one endpoints have μ_p-weights $1-p$ and p, respectively, this choice ensures that the expected error against every increasing target does not increase. We make the choices independently across coordinates, and every realization after all coordinates have been processed is increasing.

The difficult part is to retain controlled adaptive measurability. We show that the function produced by an upper shift remains measurable with respect to the old adaptive partition. Because the activations are increasing, the atom at an upper endpoint determines the atom at the corresponding lower endpoint. A lower shift requires more information. Raising the coordinate from zero to one may create new activations, but the lower atom does not record the values of the newly activated coordinates. We encode this missing information by forcing the shifted coordinate to one inside every activation. Let D be the set of coordinates on which lower shifts were chosen. For $z \in \{0,1\}^S$, let $z^{D \cap S \leftarrow 1}$ be obtained by setting the coordinates in $D \cap S$ to one. We define the family of forced activations $\mathcal{J}^D = (J_S^D)_{S \subseteq [n]}$ by $J_S^D(z) = J_S(z^{D \cap S \leftarrow 1})$ and show that the final shifted function is measurable with respect to these activations. Since each coordinate belongs to D with probability p, forcing a random D changes the effective bias from p to $p' = 2p - p^2$. For each activation involving at most d coordinates, the likelihood ratio between $\mu_{p'}$ and μ_p is at most 2^d. Writing $\mathbb{E}_D$ for expectation over the random shift choices, we obtain

$$\mathbb{E}_D \Lambda_p(\mathcal{J}^D) \le 2^d \Lambda_p(\mathcal{J}).$$

We then use a weighted averaging argument to select one realization with both small error and small active-set cost.

For the last arrow, we use monotonicity of both u and the forced activations $\mathcal{J}^D$. Fix x with $u(x) = 1$ and set every coordinate outside $A_{\mathcal{J}^D}(x)$ to zero. Every set active at x remains active because all of its coordinates are unchanged. No new set becomes active when coordinates are lowered, since the activations are increasing. The adaptive atom, and hence the value of u, is unchanged. It follows that the coordinates in $A_{\mathcal{J}^D}(x)$ at which x has value one form an exact positive certificate for u, meaning that every input that is one on these coordinates is accepted by u. We keep only certificates of size at most a cutoff w to obtain a width-w monotone DNF, and we use Markov's inequality to bound the discarded mass.

The quadratic exponent in the width bound comes from Hatami's Fourier truncation, which gives $L = \exp(O((K+1)^2/\varepsilon^2))$. For activations of arity $d = O((K+1)/\varepsilon)$, shifting and truncating the certificates give a width bound of $O(2^d L/\varepsilon)$.

Relation to prior work. Friedgut and Kalai established sharp thresholds on the additive p-scale for families invariant under a transitive permutation group [FK96]. Friedgut's junta theorem describes the dense regime [Fri98]. His later work used graph and hypergraph symmetries to show that slowly changing acceptance probabilities in sparse spaces are governed by bounded local witnesses [Fri99]. Bourgain's appendix found a bounded local conditioning that raises the acceptance probability, but not an approximation by bounded minimal elements [Fri99, Appendix]. Hatami obtained pseudo-junta approximations without symmetry assumptions and observed that, for increasing targets, it remained open whether the approximating family could be generated by small minimal elements [Hat12, Section 6]. Keller and Lifshitz work on the uniform cube, in their influence normalization, under the near-isoperimetric hypothesis

$$I(f) \le 2\mu(\log_2(1/\mu) + M), \qquad \mu = \mathbb{E}f,$$

and obtain an $\varepsilon\mu$-approximation by a DNF of size $2^{2^{O(M/\varepsilon)}}$ [KL18]. Their near-isoperimetric hypothesis and relative-error objective differ from our bounded-relative-boundary hypothesis and additive-error conclusion; our result controls width rather than DNF size. Dinur, Filmus, and Harsha obtained sparse-DNF structure under the additional assumption that the target is close to a low-degree polynomial [DFH24]. Coordinate shifting is classical and has also been used for monotone DNF approximators [BHST14]. Our contribution is a bias-matched monotonicization of Hatami's adaptive structure: the resulting function is exactly measurable with respect to a controlled forced refinement whose expected information cost remains dimension-free.

2 Hatami's pseudo-juntas

We continue with the notation from the introduction. For $i \in [n]$ and $b \in \{0,1\}$, write $x^{i\leftarrow b}$ for the point obtained by setting coordinate i to b. For $D \subseteq [n]$, define $x^{D\leftarrow 1}$ analogously by setting every coordinate in D to one.

Let $\mathcal{J} = (J_S)_{S\subseteq[n]}$ be an activation family as in (2). We write $\mathcal{F}_{\mathcal{J}} = \sigma(\Phi_{\mathcal{J}})$ for the σ-algebra generated by its adaptive partition. Thus $\mathcal{F}_{\mathcal{J}}$-measurability is precisely the measurability notion defined in the introduction. Following Hatami, an M-pseudo-junta is a Boolean function h that is $\mathcal{F}_{\mathcal{J}}$-measurable for some $\mathcal{J}$ satisfying $\mathbb{E}_p|A_{\mathcal{J}}(X)| \le M$.

Our argument uses two quantitative parameters of the activation family. We define its *arity*, denoted $\mathrm{arity}(\mathcal{J})$, as the largest $|S|$ for which $J_S \not\equiv 0$, with value zero if no such S exists. Its *multiplicity-counted activation load* is $\Lambda_p(\mathcal{J})$ from (3). These parameters control the active union through

$$|A_{\mathcal{J}}(x)| \le \sum_{S\subseteq[n]} |S| J_S(x_S), \qquad \mathbb{E}_p|A_{\mathcal{J}}(X)| \le \Lambda_p(\mathcal{J}). \tag{4}$$

The load can be much larger than the expected active-union size when active indexing sets overlap.

We use the following explicit extraction from Hatami's p-biased construction.

Lemma 2.1 (Hatami's construction with arity and load)**.** *Let $0 < p \le 1/2$, let $f : \{0,1\}^n \to \{0,1\}$, and let $0 < a \le 1$. Set $C_f = \lceil I_p(f) \rceil$, $d = \lceil 10^3 C_f/a \rceil$, and $L = \exp(10^{10} a^{-2} C_f^2)$. There are increasing activations $\mathcal{J} = (J_S)_{S\subseteq[n]}$ and a Boolean function h such that*

(i) $\mathbb{P}_p[f \ne h] \le a$;
(ii) *h is $\mathcal{F}_{\mathcal{J}}$-measurable;*
(iii) $\mathrm{arity}(\mathcal{J}) \le d$; *and*
(iv) $\Lambda_p(\mathcal{J}) \le L$.

Tracking Hatami's activation family makes its monotonicity, arity, and load guarantees explicit in the form needed below; see [Hat12, Remark 2.9 and Sections 4.3–4.4]. Hatami retains a bounded-degree family $\mathcal{S}$ of significant components in the p-biased Walsh expansion $f(x) = \sum_S F_S(x_S)$. For suitable $\eta, \delta > 0$, set $b_T = \sum_{S \in \mathcal{S}: S \supseteq T} \mathbb{P}_p[|F_S(X_S)| \geq \eta]$ and let $\mu_p^T(y) = p^{|y|}(1-p)^{|T|-|y|}$ be the point mass of $y \in \{0,1\}^T$. The activation at T is $J_T(y) = \mathbf{1}[b_T \geq \delta \mu_p^T(y)]$. The degree cutoff bounds the arity. When $p \leq 1/2$, the point mass $\mu_p^T(y)$ is nonincreasing in y, so every J_T is increasing. Moreover,

$$J_T(y) \leq \frac{b_T}{\delta \mu_p^T(y)} \quad \Longrightarrow \quad \mathbb{E}_p J_T(X_T) \leq \frac{2^{|T|} b_T}{\delta},$$

because $\mathbb{E}_p[1/\mu_p^T(X_T)] = 2^{|T|}$. Summing this estimate with multiplicity $|T|$ gives the stated load bound. Appendix A gives the details, following [Hat12, Sections 4.1–4.4].

3 Controlled monotonicization of adaptive structure

Bias-matched shifts. Our first ingredient is independent of pseudo-juntas. Fix a coordinate i. For a Boolean function v, write v_0, v_1 for its two i-slices and define

$$(U_i v)_0 = v_0, \qquad (U_i v)_1 = v_0 \vee v_1,$$
$$(L_i v)_0 = v_0 \wedge v_1, \qquad (L_i v)_1 = v_1.$$

Both shifts fix the nondecreasing fibers $00, 01, 11$. On the unique decreasing fiber 10, U_i produces 11 and L_i produces 00. OpenAI's GPT-5.6 Sol suggested trying a random choice between these shifts.

Lemma 3.1 (One-coordinate shift)**.** *Let $0 < p < 1$, let $i \in [n]$, let $f : \{0,1\}^n \to \{0,1\}$ be increasing, and let $v : \{0,1\}^n \to \{0,1\}$ be arbitrary. Choose $O_i = U_i$ with probability $1-p$ and $O_i = L_i$ with probability p, independently of the input. Then*

$$\mathbb{E}_{O_i} \mathbb{P}_p[f \neq O_i v] \leq \mathbb{P}_p[f \neq v].$$

Moreover, both $U_i v$ and $L_i v$ are nondecreasing in coordinate i, and they preserve monotonicity in every other coordinate in which v is already nondecreasing.

Proof. Condition on all coordinates other than i. The i-fiber of f is one of $00, 01, 11$. Only a 10 fiber of v changes, and the conditional errors are

fiber of f	old error	$(1-p)\,\mathrm{error}(11) + p\,\mathrm{error}(00)$
00	$1-p$	$1-p$
01	1	$(1-p)^2 + p^2$
11	p	p

The third column never exceeds the second. Integrating proves the error bound. The inequalities $v_0 \wedge v_1 \leq v_1$ and $v_0 \leq v_0 \vee v_1$ give monotonicity in coordinate i; conjunction and disjunction preserve monotonicity in every other coordinate. □

Starting from h, process the coordinates in an arbitrary fixed order, making the choices in Lemma 3.1 independently. Let $D \subseteq [n]$ be the coordinates on which the lower shift was chosen. Identifying D with its indicator vector, $D \sim \mu_p^n$; every resulting function H_D is increasing, and iterated conditional expectation gives

$$\mathbb{E}_D \mathbb{P}_p[f \neq H_D] \leq \mathbb{P}_p[f \neq h].$$

The function H_D may depend on the fixed processing order; no commutativity of the shifts is used.

The adaptive information after shifting. For $D \subseteq [n]$, define the forced activation system $\mathcal{J}^D = (J_S^D)_S$ by

$$J_S^D(z) = J_S(z^{D\cap S\leftarrow 1}),$$

where $z^{D\cap S\leftarrow 1}$ is obtained by setting the coordinates in $D \cap S$ to one. Its active union is $A_{\mathcal{J}^D}(x) = A_{\mathcal{J}}(x^{D\leftarrow 1})$. Crucially, its adaptive atom is

$$\Phi_{\mathcal{J}^D}(x) = \left(A_{\mathcal{J}}(x^{D\leftarrow 1}), x_{A_{\mathcal{J}}(x^{D\leftarrow 1})}\right).$$

The active set is computed at the forced input, and the recorded labels are those of x. When the activations are increasing, the forced partition refines the original one. Indeed, every set active at x is contained in $B = A_{\mathcal{J}^D}(x)$, so evaluating $J_S(x_S)$ for $S \subseteq B$ recovers the original active union from (B, x_B).

Lemma 3.2 (Closure under shifts)**.** *Let $i \in [n]$, let $\mathcal{J} = (J_S)_{S\subseteq[n]}$ be an activation family with increasing activations, and let $v : \{0,1\}^n \to \{0,1\}$ be $\mathcal{F}_{\mathcal{J}}$-measurable. Then $U_i v$ is $\mathcal{F}_{\mathcal{J}}$-measurable and $L_i v$ is $\mathcal{F}_{\mathcal{J}^{\{i\}}}$-measurable. Consequently, if a sequence of upper and lower shifts is applied to v, and D is the set of coordinates on which a lower shift occurs, then the final function is $\mathcal{F}_{\mathcal{J}^D}$-measurable.*

Proof. Write $v = \psi \circ \Phi_{\mathcal{J}}$. For an atom (B, a) with $i \in B$ and $a_i = 1$, let $a^{i\leftarrow 0}$ be obtained by changing the recorded i-label to zero, and put

$$\mathcal{E}_i^-(B,a) = \{S \subseteq B : J_S((a^{i\leftarrow 0})_S) = 1\}, \qquad B_i^-(B,a) = \bigcup_{S\in\mathcal{E}_i^-(B,a)} S.$$

Define

$$R_i(B,a) = (B_i^-(B,a), (a^{i\leftarrow 0})_{B_i^-(B,a)}). \tag{5}$$

When $i \notin B$, set $R_i(B,a) = (B,a)$. Any set active after lowering coordinate i was already active at the upper endpoint, since J_S is increasing. Its indexing set is therefore contained in B, and the recorded labels determine whether it remains active. If $i \notin B$ at the upper endpoint, lowering i leaves every activation unchanged. Hence $\Phi_{\mathcal{J}}(x^{i\leftarrow 0}) = R_i(\Phi_{\mathcal{J}}(x))$ whenever $x_i = 1$.

It follows that the upper shift is measurable with respect to the old partition. Define

$$\psi_U(B,a) = \begin{cases} \psi(B,a), & i \notin B, \\ \psi(B,a), & i \in B,\ a_i = 0, \\ \psi(B,a) \vee \psi(R_i(B,a)), & i \in B,\ a_i = 1. \end{cases}$$

When $i \in B$, the atom records the endpoint label, and at an upper endpoint (5) supplies the lower atom. When $i \notin B$, the shift leaves the lower endpoint unchanged; at an upper endpoint with $i \notin B$, lowering i preserves the atom. Hence $U_i v = \psi_U \circ \Phi_{\mathcal{J}}$.

For the lower shift, the upper endpoint can activate new sets containing i. The forced partition records their union while retaining the actual labels of the input. If $\Phi_{\mathcal{J}^{\{i\}}}(x) = (B,a)$, define

$$Q_i(B,a) = (B,\widehat{a}), \qquad \widehat{a}_j = \begin{cases} 1, & j = i, \\ a_j, & j \neq i, \end{cases} \quad j \in B.$$

Then $Q_i(\Phi_{\mathcal{J}^{\{i\}}}(x)) = \Phi_{\mathcal{J}}(x^{i\leftarrow 1})$. Set

$$\psi_L(B,a) = \begin{cases} \psi(Q_i(B,a)), & i \notin B, \\ \psi(Q_i(B,a)), & i \in B,\ a_i = 1, \\ \psi(R_i(Q_i(B,a))) \wedge \psi(Q_i(B,a)), & i \in B,\ a_i = 0. \end{cases}$$

If $i \notin B$, no activation containing i is active even at the upper endpoint, so the old endpoint atoms coincide. Otherwise, Q_i gives the old upper atom and $R_i \circ Q_i$ gives the old lower atom. Therefore $L_i v = \psi_L \circ \Phi_{\mathcal{J}^{\{i\}}}$.

Forcing preserves monotonicity because $z \mapsto z^{D \cap S \leftarrow 1}$ is increasing. Since $(\mathcal{J}^D)^E = \mathcal{J}^{D \cup E}$, iterating the one-coordinate result proves the assertion for a sequence of shifts. □

Remark 3.3 (Approximation on a fixed adaptive partition). The following example concerns a general family of increasing activations. At $p = 1/2$, take coordinates (b, a_1, a_2), the increasing target $f(b, a_1, a_2) = a_1 \vee a_2$, and a single nonzero activation

$$J_{\{b,a_1,a_2\}}(b, a_1, a_2) = b.$$

When $b = 0$, the four inputs form a single atom. When $b = 1$, each input is its own atom. The optimal approximator h labels the $b = 0$ atom by one and agrees with f when $b = 1$, giving error $1/8$. The lower shift satisfies $L_b h = f$, which takes both values on the $b = 0$ atom. Now consider an increasing approximator on the original partition. Labeling the $b = 0$ atom by one forces the approximator to be identically one, giving error $1/4$. Labeling it by zero gives error at least $3/8$. Thus requiring the approximator to be increasing raises the minimum error from $1/8$ to $1/4$. In Hatami's construction (Section 2), the threshold $\delta\mu_p^T(y)$ is independent of y at $p = 1/2$, so each J_T is constant.

The cost of forcing. We bound the expected size of the forced active union using the arity and activation load of the original family.

Lemma 3.4 (Load transport). *Let $0 < p < 1$, let $d \in \mathbb{Z}_{\geq 0}$, and let $\mathcal{J} = (J_S)_{S \subseteq [n]}$ be an activation family satisfying $J_S \equiv 0$ whenever $|S| > d$. Let $D \sim \mu_p^n$ and set $p' = 2p - p^2$. Writing Λ_q for the same load evaluated under bias q, we have*

$$\mathbb{E}_D \Lambda_p(\mathcal{J}^D) = \Lambda_{p'}(\mathcal{J}) \leq (2-p)^d \Lambda_p(\mathcal{J}) \leq 2^d \Lambda_p(\mathcal{J}). \tag{6}$$

Consequently,

$$\mathbb{E}_D \mathbb{E}_p |A_{\mathcal{J}^D}(X)| \leq (2-p)^d \Lambda_p(\mathcal{J}).$$

Proof. Let $X \sim \mu_p^n$ be independent of D. Each coordinate $(X^{D \leftarrow 1})_i = X_i \vee \mathbf{1}_{\{i \in D\}}$ is one with probability $p + (1-p)p = p'$, and these coordinates are independent. Thus $X^{D \leftarrow 1}$ has law $\mu_{p'}^n$. In particular, $\mathbb{E}_D \mathbb{E}_p |A_{\mathcal{J}^D}(X)| = \mathbb{E}_{p'} |A_{\mathcal{J}}(X)|$. Applying the same change of measure to each activation gives the equality in (6).

Let $|S| = s \leq d$ and let $z \in \{0,1\}^S$ have Hamming weight r. Since $p'/p = 2 - p$ and $(1-p')/(1-p) = 1 - p$,

$$\frac{\mu_{p'}^S(z)}{\mu_p^S(z)} = (2-p)^r (1-p)^{s-r} \leq (2-p)^s \leq (2-p)^d.$$

Integrating the nonnegative function J_S, multiplying by $|S|$, and summing over S proves the inequalities. The final claim follows from (4). □

We can now state and prove the controlled monotonicization theorem used in the main argument.

Theorem 3.5 (Controlled monotonicization)**.** *Let $0 < p < 1$, let $d \in \mathbb{Z}_{\geq 0}$, let $e, L \geq 0$, and let $f : \{0,1\}^n \to \{0,1\}$ be increasing. Let $\mathcal{J} = (J_S)_{S \subseteq [n]}$ be an activation family with increasing activations, and let $h : \{0,1\}^n \to \{0,1\}$ be $\mathcal{F}_{\mathcal{J}}$-measurable. Suppose $J_S \equiv 0$ for $|S| > d$ and*

$$\mathbb{P}_p[f \neq h] \leq e, \qquad \Lambda_p(\mathcal{J}) \leq L.$$

For every $\tau > 0$, there exist $D \subseteq [n]$ and an increasing Boolean function u such that

(i) *u is $\mathcal{F}_{\mathcal{J}^D}$-measurable;*

(ii) *$\mathbb{P}_p[f \neq u] \leq e + \tau$; and*

(iii) *$\mathbb{E}_p|A_{\mathcal{J}^D}(X)| \leq (2-p)^d L(1 + e/\tau)$.*

Proof. Apply the random shifts in a fixed coordinate order. For each realization D, let $e_D = \mathbb{P}_p[f \neq H_D]$ and $M_D = \mathbb{E}_p|A_{\mathcal{J}^D}(X)|$. By the shift construction, Lemma 3.2, and Lemma 3.4, every H_D is increasing and $\mathcal{F}_{\mathcal{J}^D}$-measurable, while

$$\mathbb{E}_D e_D \leq e, \qquad \mathbb{E}_D M_D \leq B := (2-p)^d L.$$

If $B = 0$, then nonnegativity gives $M_D = 0$ for every realization with positive probability; choose one with $e_D \leq e$. Otherwise,

$$\mathbb{E}_D \left[e_D + \frac{\tau}{B} M_D \right] \leq e + \tau.$$

Choose one D for which the expression inside brackets is at most its expectation. Nonnegativity gives $e_D \leq e + \tau$ and $M_D \leq B(1 + e/\tau)$. Taking $u = H_D$ proves the theorem. □

4 Certificates and completion of the proof

When the approximator u and its activations are increasing, the active coordinates at an accepted input determine a positive certificate for u.

Lemma 4.1 (The active 1-coordinates form a positive certificate)**.** *Let $\mathcal{J} = (J_S)_{S \subseteq [n]}$ be an activation family with increasing activations, and let $u : \{0,1\}^n \to \{0,1\}$ be increasing and $\mathcal{F}_{\mathcal{J}}$-measurable. For $x \in \{0,1\}^n$, set $T(x) = \{i \in A_{\mathcal{J}}(x) : x_i = 1\}$. If $u(x) = 1$, then $T(x)$ is a positive certificate for u: every $y \in \{0,1\}^n$ satisfying $y_i = 1$ for all $i \in T(x)$ has $u(y) = 1$. Consequently, for every integer $w \geq 0$,*

$$g_w(y) = \bigvee_{\substack{x \in \{0,1\}^n : \ u(x)=1 \\ |T(x)| \leq w}} \ \bigwedge_{i \in T(x)} y_i \tag{7}$$

is a monotone DNF of width at most w, satisfies $g_w \leq u$, and obeys

$$\mathbb{P}_p[u \neq g_w] \leq \mathbb{P}_p[|T(X)| > w] \leq \frac{\mathbb{E}_p|A_{\mathcal{J}}(X)|}{w+1}. \tag{8}$$

Proof. Fix x with $u(x) = 1$ and let x° be the incidence vector of $T(x)$. Equivalently, x° agrees with x on $A_{\mathcal{J}}(x)$ and is zero outside that set. We claim that every activation has the same value at x° and x. If $J_S(x_S) = 1$, then $S \subseteq A_{\mathcal{J}}(x)$, so $x^\circ_S = x_S$. If $J_S(x_S) = 0$, then $x^\circ \leq x$ and monotonicity gives $J_S(x^\circ_S) = 0$. Hence $\Phi_{\mathcal{J}}(x^\circ) = \Phi_{\mathcal{J}}(x)$, and $\mathcal{F}_{\mathcal{J}}$-measurability implies $u(x^\circ) = u(x) = 1$.

If $y_i = 1$ for every $i \in T(x)$, then $y \geq x^\circ$, so monotonicity of u gives $u(y) = 1$. Thus every term in (7) is an implicant of u, and every accepted x with $|T(x)| \leq w$ satisfies its own term. Therefore $\{x : u(x) \neq g_w(x)\} \subseteq \{x : |T(x)| > w\}$. Since $|T(X)|$ is integer-valued, Markov's inequality at threshold $w + 1$, together with $T(x) \subseteq A_{\mathcal{J}}(x)$, gives (8). □

Proof of Theorem 1.2. If f is constant, use the corresponding width-zero DNF. Otherwise, put $C_K = \lceil K \rceil$ and $a = \varepsilon/4$. Since $\lceil I_p(f) \rceil \le C_K$, Lemma 2.1 gives an approximator h and increasing activations $\mathcal{J}$ such that

$$\mathbb{P}_p[f \neq h] \le a, \qquad h \text{ is } \mathcal{F}_{\mathcal{J}}\text{-measurable}, \qquad \operatorname{arity}(\mathcal{J}) \le d, \qquad \Lambda_p(\mathcal{J}) \le L,$$

where

$$d = \left\lceil \frac{4000 C_K}{\varepsilon} \right\rceil, \qquad L = \exp\left(\frac{16 \cdot 10^{10} C_K^2}{\varepsilon^2} \right).$$

Apply Theorem 3.5 with $e = \tau = a$. Since $(2-p)^d \le 2^d$, it gives $D \subseteq [n]$ and an increasing $\mathcal{F}_{\mathcal{J}^D}$-measurable function u, where the forced activations remain increasing and

$$\mathbb{P}_p[f \neq u] \le \frac{\varepsilon}{2}, \qquad \mathbb{E}_p |A_{\mathcal{J}^D}(X)| \le 2^{d+1} L.$$

Choose $w = \lceil 4 \cdot 2^d L/\varepsilon \rceil$. Applying Lemma 4.1 to u gives a monotone width-w DNF g_w with

$$\begin{aligned} \mathbb{P}_p[f \neq g_w] &\le \mathbb{P}_p[f \neq u] + \mathbb{P}_p[u \neq g_w] \\ &\le \frac{\varepsilon}{2} + \frac{\mathbb{E}_p |A_{\mathcal{J}^D}(X)|}{w+1} \le \frac{\varepsilon}{2} + \frac{2^{d+1} L}{w+1} \le \varepsilon. \end{aligned}$$

Finally, $C_K \le K + 1$ and $0 < \varepsilon < 1$, so

$$\log w = O\left(\log \frac{1}{\varepsilon} + \frac{K+1}{\varepsilon} + \frac{(K+1)^2}{\varepsilon^2} \right) = O\left(\frac{(K+1)^2}{\varepsilon^2} \right).$$

This proves the theorem. □

Proof of Conjecture 1.1. Let $f = \mathbf{1}_{\mathcal{A}}$. By (1),

$$I_p(f) = 2(1-p)\, p \mathbb{E}_p[\operatorname{sens}_f(X)]. \tag{9}$$

If $p \le 1/2$, then $I_p(f) \le 2c$, and Theorem 1.2 applies.

Suppose $p > 1/2$. The product Poincaré inequality and (9) give

$$\operatorname{Var}_p(f) \le \frac{1}{2} I_p(f) \le c(1-p).$$

Let $\alpha = \mathbb{E}_p f$ and $m = \min\{\alpha, 1-\alpha\}$. Since $\operatorname{Var}_p(f) = m(1-m) \ge m/2$, if $1 - p \le \varepsilon/(2c)$ then $m \le \varepsilon$, and the closer constant function is an admissible width-zero approximation.

In the remaining case, $\lambda = \varepsilon/(2c) < 1 - p < 1/2$, so $p \in [\lambda, 1-\lambda]$. The relative boundary assumption directly bounds the total influence in the bit-flip convention:

$$\sum_{i=1}^{n} \mathbb{P}_p[f(X) \neq f(X^{\oplus i})] = \mathbb{E}_p[\operatorname{sens}_f(X)] \le c/p < 2c.$$

On this interval, Friedgut's biased junta theorem gives a junta h with $\mathbb{P}_p[f \neq h] \le \varepsilon$ whose coordinate set J has size bounded in terms of c and ε alone [Fri98, O'D14]. The conditional mean $m_J(z) = \mathbb{E}_p[f(X) \mid X_J = z]$ is increasing, and its Bayes threshold $\mathbf{1}[m_J(z) \ge 1/2]$ has no larger error than any other J-measurable Boolean function. Every increasing J-junta has a monotone DNF of width at most $|J|$. Taking the maximum of this bound and the low-bias bound proves the conjecture. □

Remark 4.2 (Resampling influence as $p \to 1$). A bound on $I_p(f)$ alone cannot give a width bound uniform as $p \to 1$. Let $f(x) = x_1 \wedge \cdots \wedge x_n$ and $p = 1 - 1/n$. Then

$$\mathbb{P}_p[f = 1] = p^n \longrightarrow e^{-1}, \qquad I_p(f) = 2p(1-p)np^{n-1} \longrightarrow \frac{2}{e}.$$

For fixed w, every nonempty monotone DNF of width at most w accepts with probability at least $p^w \to 1$, while the empty DNF is identically zero. Hence no fixed-width monotone DNF approximates f to sufficiently small constant error. Friedgut's all-bias quantity is instead $p\mathbb{E}_p[\mathrm{sens}_f(X)] = np^n$, which diverges in this example.

The proof gives an existential bound on width. The DNF satisfies $g_w \leq u$, and its guarantee for the original function f is the two-sided error bound $\mathbb{P}_p[f \neq g_w] \leq \varepsilon$. The optimal dependence of the width on K and ε remains open. It would also be of interest to obtain bounds on the number of terms and the computational cost of finding such a DNF.

AI methodology statement

We used OpenAI's GPT-5.6 Sol to work through the relevant literature and explore possible proof strategies. Its main contribution was to suggest using randomized coordinate shifts to make Hatami's approximator increasing (see Lemma 3.1). With assistance from OpenAI's GPT-6 Astra, we revised the exposition and completed a Lean formalization of our proofs of Theorem 1.2 and Friedgut's conjecture. We independently checked and finalized all claims and proofs and take responsibility for the paper.

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A Hatami's construction with arity and load

We prove Lemma 2.1 by tracking the activation family in Hatami's Fourier-analytic proof. We verify its arity and multiplicity-counted activation load bounds while retaining Hatami's approximation guarantee. All probabilities, expectations, and L^2-norms below are taken under μ_p^n or the appropriate coordinate marginal.

Let $f(x) = \sum_{S\subseteq[n]} F_S(x_S)$ be the p-biased Walsh expansion. In Hatami's notation, $F_S(x_S) = \widehat{f}(S)\prod_{i\in S} r_p(x_i)$, where $\widehat{f}(S)$ is the p-biased Fourier coefficient and

$$r_p(0) = -\sqrt{\frac{p}{1-p}}, \qquad r_p(1) = \sqrt{\frac{1-p}{p}}.$$

Parseval's identity and the resampling definition of influence give

$$\sum_S \|F_S\|_2^2 = \|f\|_2^2 \le 1, \qquad I_p(f) = 2\sum_S |S|\|F_S\|_2^2.$$

If $I_p(f) = 0$, full support and connectivity of the cube imply that f is constant; take $h = f$ and all activations zero. Assume henceforth that $I_p(f) > 0$, and put $C_f = \lceil I_p(f)\rceil$. Following [Hat12, Section 4], set

$$\varepsilon_0 = 10^{-3}a, \qquad k = C_f\varepsilon_0^{-1} = 10^3 a^{-1} C_f, \qquad \delta = 2^{-10^2k^2}, \qquad \varepsilon_1 = 3^{-10k^2}\varepsilon_0^{10k}.$$

Retain the Fourier sets

$$\mathcal{S} = \{S \subseteq [n] : |S| \le k,\ \|F_S\|_\infty > \varepsilon_1\}.$$

For $T \subseteq [n]$, define

$$b_T = \sum_{\substack{S\in\mathcal{S}\\ S\supseteq T}} \mathbb{P}_p[|F_S(X_S)| \ge \varepsilon_1].$$

For $|T| \le k$ and $y \in \{0,1\}^T$, set

$$J_T(y) = \mathbf{1}[b_T \ge \delta\mu_p^T(y)], \tag{10}$$

and put $J_T \equiv 0$ for $|T| > k$. This is precisely the activation family in [Hat12, Section 4.3]. The number b_T is global and independent of y. Since $p \le 1/2$, the point mass $\mu_p^T(y)$ is nonincreasing in y, so every J_T is increasing. Also, nonzero activations have arity at most $k \le d$.

It remains to record the load estimate in the form needed here. From (10), pointwise in y,

$$J_T(y) \le \frac{b_T}{\delta \mu_p^T(y)}.$$

If $X_T \sim \mu_p^T$, then full support gives

$$\mathbb{E}_p\left[\frac{1}{\mu_p^T(X_T)}\right] = \sum_{y \in \{0,1\}^T} 1 = 2^{|T|}.$$

Consequently,

$$\begin{aligned}
\Lambda_p(\mathcal{J}) &= \sum_T |T| \mathbb{E}_p J_T(X_T) \\
&\le \frac{k}{\delta} \sum_{\substack{T \subseteq [n] \\ |T| \le k}} 2^{|T|} \sum_{\substack{S \in \mathcal{S} \\ S \supseteq T}} \mathbb{P}_p[|F_S(X_S)| \ge \varepsilon_1] \\
&= \frac{k}{\delta} \sum_{S \in \mathcal{S}} \mathbb{P}_p[|F_S(X_S)| \ge \varepsilon_1] \sum_{T \subseteq S} 2^{|T|} \\
&\le \frac{k 2^{2k}}{\delta \varepsilon_1^2} \sum_{S \in \mathcal{S}} \|F_S\|_2^2 \le \frac{k 2^{2k}}{\delta \varepsilon_1^2}.
\end{aligned}$$

Here $\sum_{T \subseteq S} 2^{|T|} = 3^{|S|} \le 2^{2k}$, and the penultimate inequality is Markov's inequality. For completeness, the last parameter estimate follows directly from the chosen constants. Since $I_p(f) > 0$, we have $C_f \ge 1$, hence $k = C_f/\varepsilon_0 \ge 10^3$ and $\varepsilon_0^{-1} \le k$. Therefore

$$\frac{k 2^{2k}}{\delta \varepsilon_1^2} = k 2^{2k + 10^2 k^2} 3^{20k^2} \varepsilon_0^{-20k} \le 2^{10^3 k^2} \le \exp\left(10^{10} a^{-2} C_f^2\right) = L.$$

This is the displayed estimate on pages 518–519 of [Hat12]. Thus the same construction has the required multiplicity-counted activation load.

Finally, for exactly the activations (10), combining the estimates from Steps I–III of Hatami's proof with the final calculation in Section 4.4 yields

$$\|f - \mathbb{E}_p[f \mid \mathcal{F}_{\mathcal{J}}]\|_2^2 \le 10\varepsilon_0;$$

see equations (10)–(16) and Section 4.4 of [Hat12]. Define

$$h(x) = \mathbf{1}\left[\mathbb{E}_p[f \mid \mathcal{F}_{\mathcal{J}}](x) > \frac{1}{2}\right].$$

Then h is $\mathcal{F}_{\mathcal{J}}$-measurable. Since f is Boolean,

$$\mathbb{P}_p[f \ne h] \le 4 \|f - \mathbb{E}_p[f \mid \mathcal{F}_{\mathcal{J}}]\|_2^2 \le 40\varepsilon_0 \le a.$$

This proves Lemma 2.1.